One-Day-Ahead 1% VaR Forecasting on QQQ Daily Log Returns

A class report on rolling VaR model comparison, backtesting, and forecast uncertainty

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1 Introduction

Value-at-Risk (VaR) is one of the standard ways to summarize market downside risk. A one-day 1% VaR forecast answers the question: **conditional on information available today, how far into the left tail could tomorrow's return fall, with probability 1%?** This report studies that question for QQQ daily log returns using a rolling out-of-sample forecasting exercise.

The project compares several model families that represent different ways of forecasting tail risk:

- **parametric volatility-based VaR**: GARCH(1,1), GJR-GARCH, RiskMetrics;
- **semi-parametric empirical-tail VaR**: Filtered Historical Simulation (FHS) and residual bootstrap variants;
- **nonparametric benchmark**: Historical Simulation;
- **direct quantile modeling**: CAViaR-SAV;
- **exogenous-volatility benchmark**: package-based GARCHX(VIX).

The central empirical message is clear. For this QQQ application, the strongest practical idea is **not simply "fit a volatility model," but "fit volatility dynamically and calibrate the tail empirically."** The best overall model in the saved outputs is **GJR-FHS**, with **GARCH-FHS** a very close second. Historical Simulation remains a strong benchmark. Gaussian-tail models are systematically too optimistic in the left tail.

This report is written for classmates. The goal is therefore twofold: first, to explain what VaR forecasting is and how these models differ; second, to show what the checked repository outputs say about which methods work best in this specific one-day-ahead QQQ setting.

2 Historical Background

Modern VaR became prominent in the 1990s as financial institutions sought a simple summary of market risk over a fixed horizon and confidence level. A practical milestone was J.P. Morgan's *RiskMetrics* methodology, which popularized variance-based VaR reporting and made daily market-risk measurement more standardized. The 1996 Basel market-risk amendment then gave VaR major regulatory importance by allowing banks to use internal market-risk models, subject to quantitative backtesting requirements.

The volatility-modeling side of the story begins with Engle's ARCH model and Bollerslev's GARCH extension. These models formalized a basic empirical fact about financial returns: although raw returns often have weak serial correlation, their **volatility clusters**. Large moves tend to be followed by large moves. That feature makes dynamic variance forecasting a natural foundation for VaR.

As VaR use expanded, so did the need to validate it. Kupiec (1995) proposed the unconditional coverage test, which checks whether the observed proportion of VaR exceptions matches the nominal rate. Christoffersen (1998) added an independence criterion, emphasizing that good VaR forecasts should not only get the overall hit rate right, but also avoid clustered exceptions. Those two ideas remain the backbone of practical VaR backtesting.

Another important development was the move away from fully parametric tails. Barone-Adesi, Giannopoulos, and Vosper (1999) popularized **Filtered Historical Simulation (FHS)**, which combines a conditional volatility model with the empirical distribution of standardized residuals. This is especially appealing in financial returns because volatility can be modeled dynamically while the tail is learned from the data rather than forced into a thin Gaussian shape. Related residual-bootstrap approaches use the same spirit, but recover the predictive tail by simulation from resampled standardized residuals.

A different branch of the literature models the quantile directly instead of modeling the full conditional distribution first. Engle and Manganelli's CAViaR model is the leading example. It treats VaR itself as a dynamic object and estimates it by quantile-regression ideas. That is theoretically attractive, but in practice it requires careful optimization and adequacy testing, especially through Dynamic Quantile-style validation.

Finally, there is a natural extension of GARCH in which the variance recursion depends on an external regressor. In this project the external regressor is VIX, so the comparison benchmark is a **GARCHX(VIX)** model. This asks whether implied market volatility carries incremental information for next-day left-tail risk beyond what is already embedded in lagged QQQ returns.

This background directly motivates the final comparison in the project: not merely ARCH/GARCH versus VaR in general, but **parametric tails versus empirical tails, pure return-based volatility versus exogenous-volatility overlays, and indirect volatility-first VaR versus direct quantile recursion**.

3 Purpose Statement and Mathematical Development

3.1 Purpose

Let r_t denote the QQQ daily log return,

$$r_t = \log\left(\frac{P_t}{P_{t-1}}\right),$$

where P_t is adjusted close. The one-day-ahead 1% VaR target is the conditional lower-tail quantile satisfying

$$\Pr(r_{t+1} < \mathrm{VaR}_{t+1}(0.01) \mid \mathcal{F}_t) = 0.01.$$

Equivalently, if loss is defined as $L_{t,t+1} = -(P_{t+1} - P_t)$ in dollar space, VaR is the conditional loss threshold exceeded with probability about 1%. In this project, however, the models are implemented in return space, so the object forecasted directly is the conditional 1% quantile of r_{t+1}.

The practical aim is not in-sample fit. It is to answer a forecasting question: **which model best tracks next-day downside risk on QQQ under a realistic rolling backtest?**

3.2 Rolling forecasting design

The dataset used by the current local pipeline contains 2,824 complete QQQ/VIX observations after cleaning, running from 2015-01-05 to 2026-03-27. The saved forecast exercises use an initial training fraction of 70%, which yields

- initial training size: 1,976 observations;
- out-of-sample size: 848 one-day forecasts;
- first forecast date: 2022-11-08;
- expected number of violations at 1%: 8.48.

At each out-of-sample date t, the workflow is:

1. fit the model using data available through day $t-1$;
2. construct a one-day-ahead VaR forecast for day t;
3. observe realized return r_t;
4. record the hit indicator
$$I_t = \mathbf{1}\{r_t < \widehat{\mathrm{VaR}}_t\};$$
5. backtest the hit sequence.

This is the right design for a risk report because a VaR model should be judged by **forecast calibration**, not by how well it fits the past sample in hindsight.

3.3 Common modeling form

Many of the models can be written in conditional location-scale form,

$$r_{t+1} = \mu_{t+1} + \sigma_{t+1} z_{t+1},$$

which implies

$$\mathrm{VaR}_{t+1}(\alpha) = \mu_{t+1} + \sigma_{t+1} q_\alpha.$$

The key difference across models is therefore how they determine:

- the conditional scale σ_{t+1},
- the tail quantile q_α.

That decomposition helps organize the model comparison.

3.4 Models in the checked project workflow

3.4.1 GARCH(1,1): parametric volatility baseline

The standard GARCH(1,1) model uses

$$\varepsilon_t = r_t - \mu,$$

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2.$$

In the main Python runner, the saved GARCH baseline uses Student-t innovations. Thus the one-step VaR forecast is

$$\widehat{\mathrm{VaR}}_{t+1}(\alpha) = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} q_{\alpha,\,t}^{(t)},$$

where $q_{\alpha,\,t}^{(t)}$ is the fitted standardized Student-t quantile.

This model is useful because it is a clean parametric benchmark: dynamic volatility, but fully distribution-based tail mapping.

3.4.2 GJR-GARCH: leverage-sensitive parametric baseline

To allow asymmetric response to bad news, GJR-GARCH uses

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \gamma \varepsilon_{t-1}^2 I(\varepsilon_{t-1} < 0) + \beta \sigma_{t-1}^2.$$

If $\gamma > 0$, negative returns increase future volatility more than positive returns of the same magnitude. That is empirically plausible for equity returns.

3.4.3 Historical Simulation: nonparametric benchmark

Historical Simulation uses the empirical lower-tail quantile of the most recent W returns,

$$\widehat{\mathrm{VaR}}_{t+1}(\alpha) = \mathrm{Quantile}_{\alpha}(r_{t-W+1}, \dots, r_t).$$

In the saved code, $W = 250$. This method is attractive because it is transparent and avoids explicit distributional assumptions. Its tradeoff is slower adaptation when volatility shifts sharply.

3.4.4 Filtered Historical Simulation (FHS): empirical-tail semi-parametric VaR

FHS first estimates a volatility model and then applies the empirical quantile of standardized residuals:

$$\widehat{\mathrm{VaR}}_{t+1}(\alpha) = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1}\hat{q}_{\alpha,\mathrm{emp}}.$$

This is the central methodological winner in the project. The volatility dynamics are modeled parametrically, but the left-tail innovation quantile is taken from the empirical residual distribution. In other words, FHS keeps the dynamic volatility structure of GARCH while relaxing the final tail assumption.

The saved project contains both Gaussian-based and Student-t-based FHS variants:

- GARCH-FHS,
- GJR-FHS,
- GARCH-t-FHS,
- GJR-t-FHS.

3.4.5 Residual bootstrap VaR: simulation-based empirical tails

The bootstrap variants resample standardized residuals, simulate one-step returns, and then take the empirical 1% quantile of the simulated distribution:

$$r_{t+1}^{*(i)} = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} z_{t+1}^{*(i)},$$

with the VaR estimate defined as the empirical α-quantile across the $r_{t+1}^{*(i)}$ draws.

For one-step forecasting, this is mechanically very close to FHS. That is why the FHS and bootstrap variants look similar in the saved outputs.

3.4.6 CAViaR-SAV: direct quantile dynamics

The current implementation uses the Symmetric Absolute Value form of CAViaR. The code evolves the positive VaR magnitude by

$$v_t = \beta_0 + \beta_1 v_{t-1} + \beta_2 |r_{t-1}|,$$

and then maps back to the left-tail quantile via

$$q_t = -v_t.$$

Parameters are estimated by minimizing the quantile check loss

$$L(\beta) = \sum_t \rho_\alpha(r_t - q_t(\beta)),$$

where

$$\rho_\alpha(u) = u(\alpha - I(u < 0)).$$

Conceptually, this is appealing because it models the quantile directly. In the current project, though, it should be treated as **exploratory rather than fully mature**: only the SAV form is implemented, optimization is light, only a few starting values are used, and the paper's Dynamic Quantile-style adequacy machinery is not implemented.

3.4.7 RiskMetrics: lecture-aligned benchmark

RiskMetrics uses an exponentially weighted variance recursion,

$$\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_{t-1}^2,$$

with $\lambda = 0.94$ in the saved benchmark. This is a classic benchmark because it is historically important and easy to explain, but it imposes a Gaussian tail mapping.

3.4.8 GARCHX(VIX): exogenous-volatility benchmark

The updated benchmark is implemented separately in `GARCHX.r` using the R `garchx` package. In each rolling window, returns are demeaned and VIX is standardized, giving the benchmark structure

$$y_t = r_t - \mu,$$

$$\sigma_t^2 = \omega + \alpha y_{t-1}^2 + \beta \sigma_{t-1}^2 + \delta x_t,$$

where x_t is standardized VIX. The current script then maps volatility into VaR by a Gaussian quantile,

$$\widehat{\mathrm{VaR}}_{t+1}(\alpha) = \mu + \hat{\sigma}_{t+1|t} \Phi^{-1}(\alpha).$$

This is best interpreted as a **package-based Gaussian-QML GARCHX comparison benchmark**. It is cleaner and more defendable than the old custom Python implementation, but it is still not an empirical-tail method.

4 Data Example with Representative Code

4.1 The actual data setting

The project uses:

- QQQ adjusted close prices;
- VIX close values;
- QQQ daily log returns as the target series;
- VIX only for the GARCHX benchmark.

A minimal data-preparation sketch is:

```
import pandas as pd

df = pd.read_csv("data/model_data.csv", parse_dates=["Date"])
df = df[["Date", "log_ret", "vix_close"]].dropna().sort_values("Date").reset_index(drop=True)

alpha = 0.01
initial_train = 0.7
n = len(df)
train_end0 = int(n * initial_train)
```

The essential modeling idea is that at each date, only the data available up to that point are used to estimate the next-day left-tail quantile.

4.2 A worked one-day example

Table 1 shows a representative date from the saved outputs: 2025-04-04, a day with a large negative QQQ return. This date is useful because it reveals how the models differ when stress is real rather than hypothetical.

Date	Model	Forecasted VaR	Realized return	Violation?
2025-04-04	GJR-FHS	-0.07956	-0.06412	0
2025-04-04	GARCH-FHS	-0.07059	-0.06412	0
2025-04-04	GARCH(1,1)	-0.06311	-0.06412	1
2025-04-04	Historical-Simulation	-0.03664	-0.06412	1
2025-04-04	RiskMetrics (0.94)	-0.04525	-0.06412	1
2025-04-04	GARCHX(VIX) [garchx pkg]	-0.04914	-0.06412	1
2025-04-04	CAViaR-SAV	-0.06293	-0.06412	1

This single date illustrates the project's main conclusion. The empirical-tail FHS models move far enough into the left tail to avoid a violation, whereas the Gaussian-style and slower-moving alternatives generally do not. Of course, one date is not a backtest, but it helps make the model differences concrete.

4.3 Representative rolling-forecast code

The following simplified code sketch captures the project workflow:

```
rows = []
for t in range(train_end0, n):
    train = df.iloc[:t]
    test_row = df.iloc[t]

    r_train = train["log_ret"].to_numpy()
    realized = float(test_row["log_ret"])

    var_t = fitted_model_forecast(r_train, alpha=0.01)
    hit_t = int(realized < var_t)

    rows.append({
        "Date": test_row["Date"],
        "VaR": var_t,
        "Return": realized,
        "Violation": hit_t,
    })
```

That loop is conceptually the heart of the report. Every backtest table and every VaR-versus-return plot comes from repeating this one-step logic across the full out-of-sample period.

4.4 Workflow provenance and what each script contributes

Because the final project evolved over time, the saved results come from **multiple runners**, and that provenance should be stated explicitly.

- `run_var_project.py` produces the main saved files `outputs/var_forecasts.csv` and `outputs/var_backtests.csv`. These contain **seven models**:
  - GARCH(1,1)
  - GJR-GARCH
  - CAViaR-SAV
  - Historical-Simulation
  - GARCH-t-FHS
  - GJR-t-FHS
  - GARCH-t-Bootstrap
- `run_var_ext.py` produces `outputs/var_forecasts_ext.csv` and `outputs/var_backtests_ext.csv`. These contain **four extension models**:
  - GARCH-Gaussian
  - GARCH-FHS
  - GJR-FHS
  - GARCH-Bootstrap
- `run_var_riskmetrics.py` produces the separate benchmark files under `outputs/riskmetrics/`.
- `GARCHX.r` produces the package-based benchmark files under `outputs/garchx_only/`.

So the final report's cross-model comparison is a **manually synthesized comparison across saved outputs from separate runners**, all using the same out-of-sample horizon of 848 days. The package-based GARCHX benchmark is therefore reported as a separate checked benchmark series, not as a model silently embedded inside the Python integrated output files.

5 Empirical Results

5.1 Main backtest evidence

Table 2 summarizes the checked one-day-ahead 1% VaR backtests across the saved outputs. The table combines the main runner, the extension runner, and the two standalone benchmark scripts. Because the out-of-sample size is the same in all of them ($N = 848$), the comparison is directly interpretable.

Model	Source	Violations	Rate	Kupiec p	Christoffersen p	Conditional coverage p
GJR-FHS	`run_var_ext.py`	8	0.94%	0.8672	0.6961	0.9137
GARCH-FHS	`run_var_ext.py`	7	0.83%	0.5984	0.7327	0.8212

Model	Source	Violations	Rate	Kupiec p	Christof-fersen p	Conditional coverage p
GARCH-t-FHS	`run_var_project.py`	6	0.71%	0.3664	0.7698	0.6371
GJR-t-FHS	`run_var_project.py`	6	0.71%	0.3664	0.7698	0.6371
GARCH-t-Bootstrap	`run_var_project.py`	6	0.71%	0.3664	0.7698	0.6371
Historical-Simulation	`run_var_project.py`	12	1.42%	0.2529	0.1569	0.1910
GARCH(1,1)	`run_var_project.py`	13	1.53%	0.1480	0.1890	0.1482
CAViaR-SAV	`run_var_project.py`	6	0.71%	0.3664	0.0297	0.0625
GJR-GARCH	`run_var_project.py`	15	1.77%	0.0424	0.4621	0.0972
GARCHX(VIX) [garchx pkg]	`GARCHX.r`	15	1.77%	0.0424	0.2621	0.0679
GARCH-Bootstrap	`run_var_ext.py`	5	0.59%	0.1934	0.8075	0.4166
RiskMetrics ($\lambda = 0.94$)	`run_var_riskmetrics.py`	18	2.12%	0.0043	0.3925	0.0117
GARCH-Gaussian	`run_var_ext.py`	22	2.59%	0.0001	0.5970	0.0005

Three broad patterns stand out.

5.1.1 1. The clean headline result: GJR-FHS is best, GARCH-FHS is a close second

If the goal is a clear and interpretable final recommendation, the two strongest practical models are **GJR-FHS** and **GARCH-FHS**.

- **GJR-FHS** is closest to the nominal 1% violation target and passes the unconditional coverage, independence, and conditional coverage checks comfortably.
- **GARCH-FHS** is almost as strong, with a slightly more conservative hit rate.

These are the strongest final recommendations because they combine good statistical performance with a simple, defensible story: dynamic volatility forecast plus empirical tail calibration.

5.1.2 2. Empirical-tail calibration matters more than simply changing the volatility recursion

The Student-*t* empirical-tail variants from the main runner also perform very well, and their backtest p-values place them near the top mechanically. That is worth acknowledging directly. However, the cleaner final class-report story is that the winning idea is **empirical-tail calibration itself**, with GJR-FHS and GARCH-FHS being the most interpretable leading examples.

By contrast, simply moving from GARCH to GJR-GARCH without empirical-tail calibration does not solve the problem. GJR-GARCH still produces too many violations, so leverage alone is not enough if the tail mapping remains too restrictive.

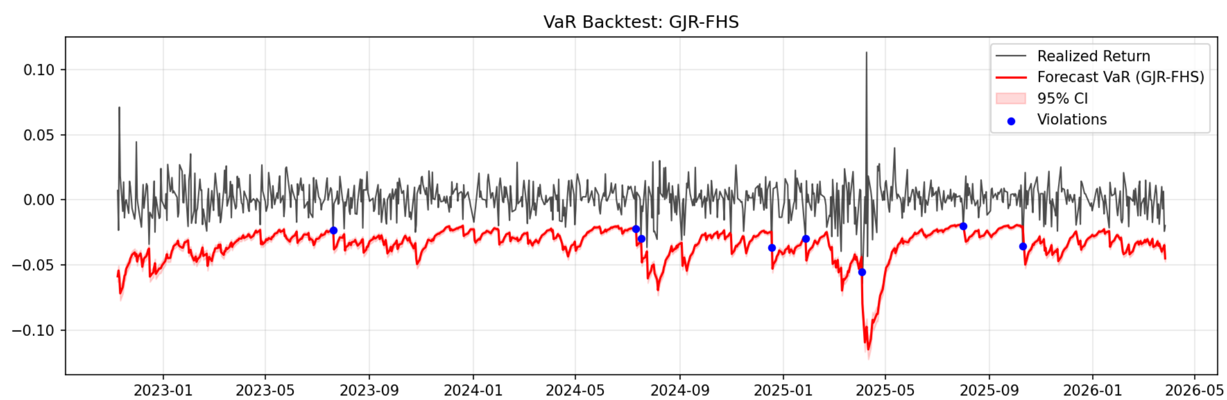
5.1.3 3. Gaussian tails fail for this dataset

RiskMetrics and GARCH-Gaussian both produce too many violations. Their issue is not that the volatility recursion is useless; it is that the final left-tail mapping is too thin for QQQ downside risk. In this dataset, the lower tail is too heavy for a Gaussian quantile to provide a reliable 1% cutoff.

5.2 What the plots add

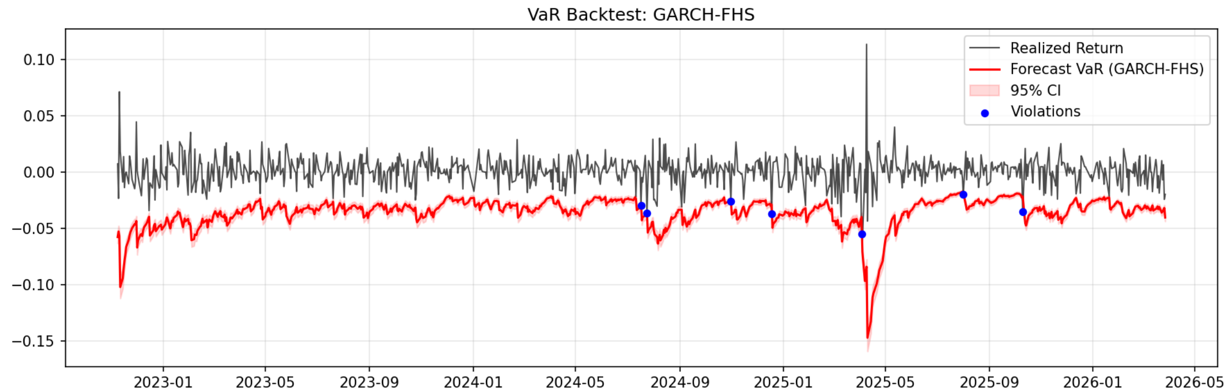
The saved plots are helpful because they show **how** the models behave during stress rather than only how many exceptions they produce overall.

5.2.1 Best overall model: GJR-FHS



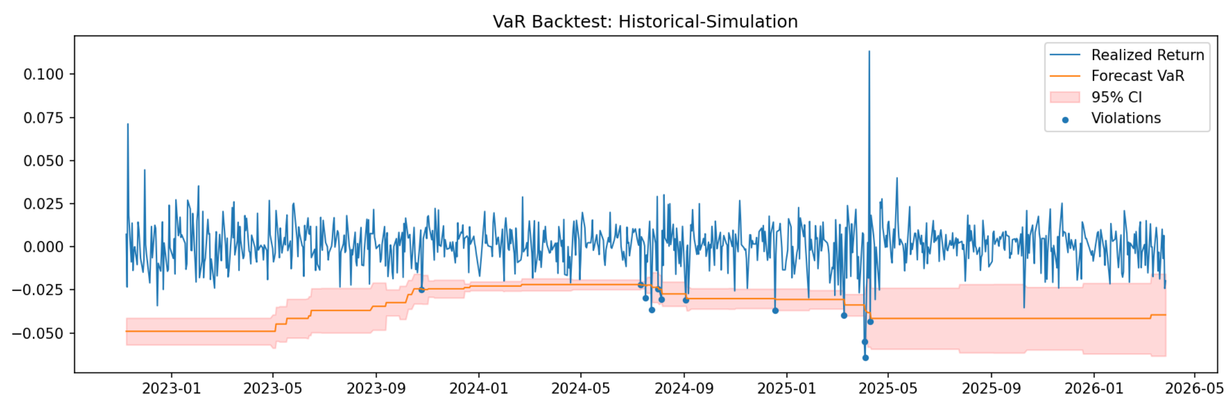
The GJR-FHS VaR line moves sharply downward in turbulent periods, but it does not remain excessively conservative once stress eases. The plot therefore supports the backtest table: the model reacts strongly enough when risk rises, without staying permanently too far in the tail.

5.2.2 Close second: GARCH-FHS



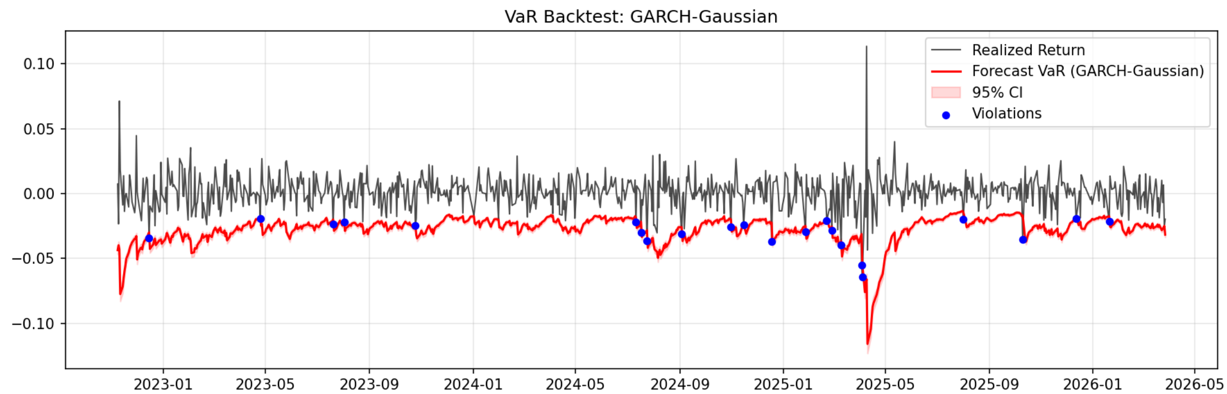
GARCH-FHS shows almost the same practical story. Relative to GJR-FHS, it is slightly less flexible about asymmetry, but its overall calibration is still excellent.

5.2.3 Strong benchmark: Historical Simulation



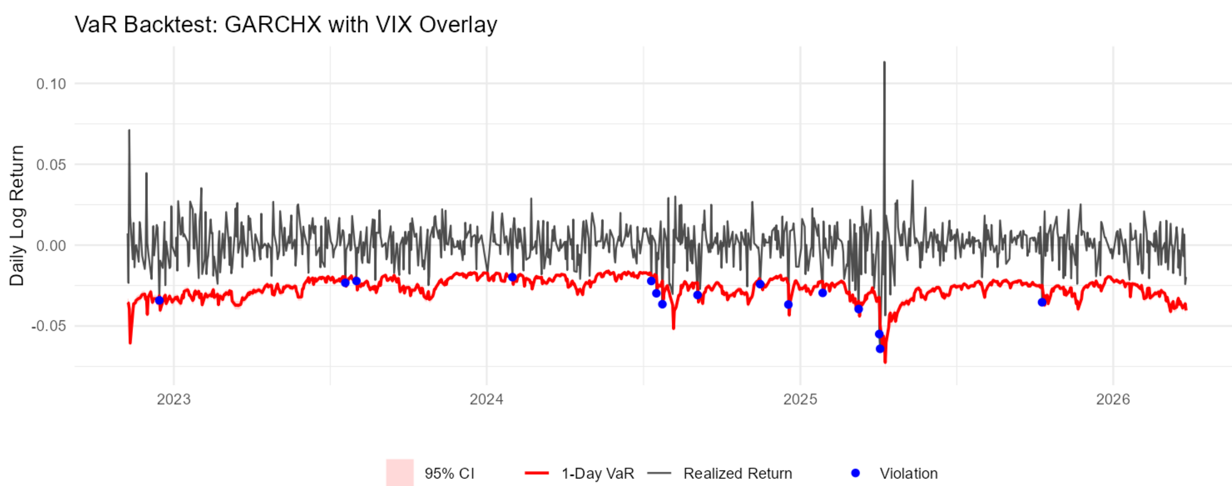
Historical Simulation is visibly more stepwise and less adaptive. That slower movement explains why it misses some stress episodes more easily than the best FHS models. Even so, its backtests remain respectable, which is why it deserves a place in the final comparison.

5.2.4 Why Gaussian models lose



The Gaussian GARCH line sits too close to zero during volatile periods. The plot makes the backtest result intuitive: too many realized losses fall below the VaR line because the Gaussian tail is simply not deep enough.

5.2.5 Exogenous-volatility benchmark



The package-based GARCHX benchmark is methodologically cleaner than the old custom implementation, but it still does not match the best empirical-tail models. The plot suggests a coherent volatility forecast, yet the Gaussian tail conversion remains too shallow in practical left-tail calibration.

5.3 Sharpened model positioning

The checked evidence supports the following final positioning.

- **Main recommendation:** GJR-FHS
- **Close second:** GARCH-FHS
- **Strong benchmark:** Historical Simulation
- **Useful parametric baseline:** GARCH(1,1)
- **Comparison benchmark with exogenous information:** package-based GARCHX(VIX)
- **Exploratory direct-quantile model:** CAViaR-SAV

The Student-t empirical-tail variants also deserve honest acknowledgement as excellent performers. But for a clean and teachable final class report, GJR-FHS and GARCH-FHS are the most natural headline models because they are strong, intuitive, and easy to explain.

6 Forecast Uncertainty and Confidence Interval Interpretation

The assignment requires some measure of forecast uncertainty, and the updated project materials provide exactly that through the saved `lower_bound` and `upper_bound` columns in the forecast files. Those bounds make it possible to discuss not only point VaR forecasts, but also how uncertain those point forecasts are.

6.1 Why uncertainty matters in VaR forecasting

A point VaR forecast is itself an estimate of a conditional quantile. It therefore inherits uncertainty from several sources:

- finite-sample estimation error;
- volatility-model estimation error;
- uncertainty in the tail quantile;
- model misspecification.

In the updated presentation, slides 27–29 explain this in terms of 95% confidence intervals for VaR. A generic asymptotic quantile interval takes the form

$$\hat{q}_\alpha \pm 1.96\,\mathrm{SE}(\hat{q}_\alpha),$$

where, in large samples,

$$\mathrm{SE}(\hat{q}_\alpha) = \sqrt{\frac{\alpha(1-\alpha)}{n\,f(\hat{q}_\alpha)^2}}.$$

For conditional location-scale VaR models,

$$\mathrm{VaR}_{t+1}(\alpha) = \mu_{t+1} + \sigma_{t+1} q_\alpha,$$

so quantile uncertainty is mapped into uncertainty for the final VaR forecast.

6.2 Why the intervals are not perfectly apples-to-apples

One important caution is that the saved intervals are **not constructed in exactly the same way for every model family**.

- **Parametric Gaussian / Student-t models** use asymptotic quantile-standard-error logic pushed through the location-scale mapping.
- **FHS models** estimate the volatility parametrically but take the tail quantile from the empirical residual distribution, so the interval reflects uncertainty in that empirical tail estimate.
- **Bootstrap models** simulate one-step returns and take interval information directly from the simulated predictive distribution.
- **Historical Simulation** uses a rolling empirical quantile interval from the raw return window.
- **CAViaR-SAV** uses a centered empirical proxy interval rather than a full inferential interval from a fitted quantile-dynamics likelihood or Hessian.

So interval widths are informative, but they should be read as **model-family-specific measures of VaR uncertainty**, not as perfectly identical statistical objects.

6.3 Comparing interval widths across the saved outputs

A useful summary statistic is the average interval width

$$\text{width}_t = \text{upper\_bound}_t - \text{lower\_bound}_t.$$

Using the current saved forecast files, the average widths are:

Model	Mean interval width
CAViaR-SAV	0.02064
Historical-Simulation	0.02064
GARCH-t-FHS	0.00783
GARCH-FHS	0.00688
GARCH(1,1)	0.00677
GJR-GARCH	0.00637
GJR-t-FHS	0.00586
GJR-FHS	0.00436
GARCH-Gaussian	0.00369
RiskMetrics ($\lambda = 0.94$)	0.00368
GARCH-t-Bootstrap	0.00368
GARCHX(VIX) [garchx pkg]	0.00353
GARCH-Bootstrap	0.00303

Several interpretations are useful.

6.3.1 1. Historical Simulation and CAViaR-SAV are the least precise

Historical Simulation and CAViaR-SAV have the widest average intervals by a large margin. For Historical Simulation this makes sense: an extreme empirical quantile estimated from a rolling 250-day window is inherently noisy. For CAViaR, the wide interval reflects both quantile uncertainty and the fact that the current implementation uses a proxy interval construction rather than a tightly specified inferential one.

In practical terms, those models produce VaR thresholds that are less precisely pinned down day by day.

6.3.2 2. The best FHS models are both calibrated and reasonably stable

GJR-FHS and GARCH-FHS are attractive not because they have the narrowest intervals in absolute terms, but because they combine:

- strong backtesting performance,
- moderate interval width,
- interpretable dynamics.

That combination is more meaningful than raw narrowness alone.

6.3.3 3. Narrow intervals can still be misleading if the model is badly calibrated

GARCH-Gaussian, RiskMetrics, and the package-based GARCHX benchmark all have comparatively tight intervals. But they also perform worse in backtests than GJR-FHS and GARCH-FHS. The lesson is simple: **a narrow confidence band around a biased or thin-tailed VaR forecast is not a practical advantage.**

6.3.4 4. Bootstrap intervals are tight partly because of construction

The bootstrap variants produce relatively small average widths. That is not surprising for one-step forecasting, where bootstrap and FHS are already very close. The saved bootstrap implementation also resets the random seed within the rolling forecast function, so the bootstrap should be interpreted as reproducible but somewhat mechanically constrained.

6.4 Practical interpretation for risk forecasting

A wider VaR interval means the location of the next-day 1% tail threshold is itself more uncertain. A narrower interval means the point estimate is more concentrated **within that model's own construction**. But the useful classroom lesson is this:

> the most persuasive VaR model is not the one with the narrowest interval; it is the one that combines reasonable interval width with good backtest calibration.

On that standard, the uncertainty analysis reinforces rather than overturns the project's headline result: **GJR-FHS and GARCH-FHS are the most convincing practical choices.**

7 Discussion

The main lesson of the project is that **empirical-tail calibration beats purely parametric tail mapping** in this QQQ one-day-ahead 1% VaR setting. That conclusion is supported by both the backtest tables and the saved VaR plots.

First, the project shows that dynamic volatility alone is not enough. GARCH and GJR-GARCH are useful volatility models, but if they are paired with a thin or overly rigid tail assumption, their VaR forecasts still miss too often. The problem is therefore not only variance forecasting; it is tail forecasting.

Second, asymmetry helps, but the biggest gain comes from the residual-tail treatment. This is why GJR-FHS edges out GARCH-FHS, yet both clearly outperform their more purely parametric counterparts. The key improvement is the FHS structure itself.

Third, Historical Simulation remains important. It is slower and less adaptive, but it is interpretable, assumption-light, and surprisingly competitive. That is a healthy reminder that more elaborate models need to earn their complexity.

Fourth, the exogenous-volatility benchmark is useful conceptually even though it is not the final winner. VIX is economically meaningful, and testing GARCHX(VIX) is a sensible modeling question. But the checked saved outputs do not show enough benefit to beat the best empirical-tail return-based models.

Finally, the CAViaR discussion is more nuanced than in earlier project notes. The saved outputs do **not** support calling CAViaR-SAV a catastrophic failure. Its unconditional hit rate is actually conservative and acceptable. The main issue is instead **exception clustering** and the fact that the implementation is only partial. That is a more accurate and fair interpretation of what the saved results show.

8 Limitations

This final report is stronger than a project memo, but the project still has limitations that should be stated clearly.

1. **Split workflow.** The final comparison draws from multiple saved runners rather than one integrated master script. The comparison is still valid because the out-of-sample horizon is aligned at 848 days, but the workflow is less tidy than an all-in-one pipeline.
2. **GARCHX benchmark status.** The current exogenous-volatility benchmark is cleaner because it is package-based, but it is still separate from the Python integrated files and still uses a Gaussian VaR overlay. So it should be viewed as a comparison benchmark, not the final recommended model.
3. **CAViaR implementation scope.** The project only implements the SAV specification, uses light optimization, and omits fuller adequacy machinery such as Dynamic Quantile-style validation. So the CAViaR result is informative, but not a full replication of the original method.

4. **Bootstrap implementation detail.** The current bootstrap code resets the random seed inside the forecast function, which makes the rolling bootstrap deterministic across iterations. That improves reproducibility but limits the spirit of a genuinely varying bootstrap exercise.
5. **Application scope.** The conclusions are specific to one asset, one horizon, and one quantile level. The ranking could change for other assets, multi-day horizons, or Expected Shortfall.

9 Conclusion

This project studies one-day-ahead 1% VaR forecasting for QQQ daily log returns under a rolling out-of-sample design with 848 forecast dates. The checked repository evidence supports a clear final class-report conclusion.

The most successful models are those that combine **dynamic volatility estimation with empirical tail calibration**. The strongest practical recommendation is **GJR-FHS**, with **GARCH-FHS** as a close second. Historical Simulation remains a strong benchmark, and GARCH(1,1) remains a useful parametric baseline. By contrast, Gaussian-tail approaches such as GARCH-Gaussian and RiskMetrics underestimate downside risk in this application.

The package-based **GARCHX(VIX)** benchmark is worth including because it asks whether exogenous market volatility improves the forecast. In the saved outputs, however, it does not outperform the best empirical-tail models. The exploratory **CAViaR-SAV** result is also worth reporting honestly: it no longer looks like a coverage disaster, but it remains limited by exception clustering and partial implementation.

The uncertainty analysis strengthens the same story. Wider or narrower interval widths are informative, but they are not identical objects across all model families. The most persuasive models are not simply those with the narrowest intervals, but those with both credible uncertainty behavior and strong backtest calibration. On that standard, **GJR-FHS and GARCH-FHS remain the best final choices**.

The clean final takeaway is therefore this: for one-day-ahead 1% VaR on QQQ, the best forecasting story is not "choose the fanciest volatility model," but rather **forecast volatility dynamically and let the data speak more flexibly about the tail**.

10 References

Barone-Adesi, G., Giannopoulos, K., & Vosper, L. (1999). *VaR without correlations for portfolios of derivative securities.* Journal of Futures Markets, 19(5), 583–602.

Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics, 31*(3), 307–327.

Christoffersen, P. F. (1998). Evaluating interval forecasts. *International Economic Review, 39*(4), 841–862.

Engle, R. F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica, 50*(4), 987–1007.

Engle, R. F., & Manganelli, S. (2004). CAViaR: Conditional autoregressive value at risk by regression quantiles. *Journal of Business & Economic Statistics, 22*(4), 367–381.

J.P. Morgan/Reuters. (1996). *RiskMetrics technical document* (4th ed.). New York: J.P. Morgan.

Kupiec, P. H. (1995). Techniques for verifying the accuracy of risk measurement models. *Journal of Derivatives, 3*(2), 73–84.

Pascual, L., Romo, J., & Ruiz, E. (2006). Bootstrap prediction for returns and volatilities in GARCH models. *Computational Statistics & Data Analysis, 50*(9), 2293–2312.